Plan 2, the core

What each piece controls, and how they assemble

Setup

We want to prove $\mathcal{A}(\Omega)^2 \le C\,\delta_T(\Omega)$ (the Saint–Venant version of stability). Kohler–Jobin and bounded reduction will then upgrade this to the Faber–Krahn inequality with the right rate.

Tools we use:

- the **torsion function** u_Ω: $-\Delta u = 1$ in Ω, $u = 0$ on $\partial\Omega$;
- the **superlevel sets** $E_\rho = \{u_\Omega > t(\rho)\}$, parametrised so that $|E_\rho| = \omega_n \rho^n$, $\rho \in [0,1]$;
- the **rescaled sets** $F_\rho = \rho^{-1} E_\rho$, all of volume $|B_1| = \omega_n$;
- the metric space $(\mathcal{X}, d_{\mathcal{X}})$ of finite-perimeter sets modulo translation, with $d_{\mathcal{X}}([A],[B]) = \inf_v |A \triangle (B+v)|$.

As ρ runs from ρ_* (small) up to 1, the family $[F_\rho]$ is a path in $\mathcal{X}$. Our target ball B_1 also lives in $\mathcal{X}$. We will show that this path passes within distance $\sqrt{\delta_T}$ of $[B_1]$ at a specific $\widehat{\rho}$ near 1, and then transfer the closeness back to Ω.

1. The deficit identity: two defects per level

The starting fact is an identity (not just an inequality) that breaks the global deficit into a ρ-integral of two non-negative pointwise quantities:

$$\boxed{\delta_T(\Omega) \;=\; \int_{\rho_*}^{1} \big(D_H(\rho) + D_I(\rho)\big)\, w(\rho)\, d\rho,}$$

with explicit weight $w(\rho) > 0$ and

$$D_H(\rho) \;:=\; \alpha(\rho)\beta(\rho) - P(E_\rho)^2, \qquad D_I(\rho) \;:=\; P(E_\rho)^2 - c_n |E_\rho|^{2-2/n},$$

where $\alpha = \int_{\partial^* E_\rho} |\nabla u|^{-1}$ and $\beta = \int_{\partial^* E_\rho} |\nabla u|$. Both are ≥ 0 (Cauchy–Schwarz / isoperimetric). The identity comes from refusing to discard either defect inside Talenti's symmetrisation proof of $T(\Omega) \le T(B^*)$.

Consequence. Both

$$\int D_H\, w\, d\rho \;\le\; \delta_T, \qquad \int D_I\, w\, d\rho \;\le\; \delta_T.$$

What each defect *is*, at one level:

- $D_I(\rho)$ measures how **non-ball** the level set E_ρ itself is. $D_I = 0$ iff E_ρ is a ball.
- $D_H(\rho)$ measures how **non-uniformly** $|\nabla u|$ distributes on $\partial^* E_\rho$. $D_H = 0$ iff $|\nabla u|$ is constant on the level set.

These are the only two ingredients we extract from δ_T. Plan 2 uses each for one specific job. **We cannot do either job without the other.** The next two sections explain what each controls and the section after that explains why both are needed simultaneously.

2. D_I controls how far each level is from a ball (*potential*)

For each ρ where $D_I(\rho)$ is below a fixed threshold (these are the **good levels**; the bad ones have small total μ-measure by Markov, so we ignore them), Fusco–Julin's quantitative isoperimetric inequality says: there is a centre $z_\rho \in \mathbb{R}^n$ such that

$$|E_\rho \triangle B_\rho(z_\rho)|^2 \;\le\; \rho^{2n} \cdot C\, D_I(\rho).$$

Rescaling by ρ^{-1}:

$$\boxed{d_{\mathcal{X}}([F_\rho],[B_1])^2 \;\le\; C\, D_I(\rho).}$$

In words: $D_I(\rho)$ **is the pointwise distance^2 from $[F_\rho]$ to $[B_1]$ in $\mathcal{X}$**, up to constants.

Think of $d_{\mathcal{X}}([F_\rho],[B_1])^2$ as a **potential energy** for the path $[F_\rho]$ in $\mathcal{X}$: it tells us, at each ρ, how far the path currently is from the target B_1. Integrating against $w\,d\rho$:

$$\int d_{\mathcal{X}}([F_\rho],[B_1])^2\, w\, d\rho \;\le\; C \int D_I\, w\, d\rho \;\le\; C\,\delta_T.$$

Could we just use this? No. "$\int$(potential) $\le \delta_T$" means the path is on average near B_1 — but we need a *specific* ρ where the path is provably near B_1, and that specific ρ has to be close to 1 (so the level set is close to Ω). A pure measure statement does not let us locate a specific good ρ. For that we need to know the path can't dart in and out of a neighbourhood of B_1 — i.e., a control on its *speed*.

3. D_H controls how fast each level moves (*kinetic*)

The path $\rho \mapsto [F_\rho]$ in $\mathcal{X}$ has a well-defined metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$ (absolutely continuous in ρ). Its size is determined by the normal velocity of the boundary $\partial^* E_\rho$ under ρ. The normal velocity is

$$V_\rho(x) \;=\; \frac{-t'(\rho)}{|\nabla u(x)|}, \qquad x \in \partial^* E_\rho.$$

After accounting for the rescaling $F_\rho = \rho^{-1} E_\rho$, the speed in $\mathcal{X}$ comes out to

$$\boxed{|\dot{F}_\rho|_{\mathcal{X}}^2 \;\le\; C\, D_H(\rho) \cdot (\text{explicit weight}).}$$

Why D_H? Because $|\dot{F}_\rho|_{\mathcal{X}}$ is sensitive only to the *non-uniform* part of the motion of the boundary: a level set that moves uniformly outward (constant V_ρ) just rescales, and is invisible in $\mathcal{X}$ (where we mod out by translation and the volume is fixed by the parametrisation). The non-uniformity of V_ρ is exactly the non-uniformity of $|\nabla u|$ on $\partial^* E_\rho$, which is what D_H measures.

Integrating:

$$\int |\dot{F}_\rho|_{\mathcal{X}}^2\, w\, d\rho \;\le\; C \int D_H\, w\, d\rho \;\le\; C\,\delta_T.$$

In words: **the kinetic energy of the path $[F_\rho]$ in $\mathcal{X}$ is bounded by δ_T.** If δ_T is small, the path moves slowly.

4. Why we need both: the abstract metric trace

Now combine. From Sections 2 and 3:

$$\boxed{\int_{\rho_*}^{1} \Big(\underbrace{d_{\mathcal{X}}([F_\rho],[B_1])^2}_{\text{potential}} + \underbrace{|\dot{F}_\rho|_{\mathcal{X}}^2}_{\text{kinetic}} \Big)\, w\, d\rho \;\le\; C\,\delta_T(\Omega).}$$

This is the integrated "action" of the path $[F_\rho]$.

The abstract weighted metric trace lemma. In any metric space, if an absolutely continuous path $\gamma : [a,b] \to \mathcal{X}$ satisfies

$$\int_a^b \big(d(\gamma(\rho),p)^2 + |\dot\gamma(\rho)|^2\big)\, w(\rho)\, d\rho \;\le\; \varepsilon,$$

then on any sub-window $[c, c+L]$ with weighted length L there exists a specific point $\widehat{\rho} \in [c, c+L]$ with

$$d(\gamma(\widehat{\rho}),p)^2 \;\le\; \frac{C\varepsilon}{L}.$$

The proof is averaging plus the triangle inequality with the kinetic bound: *Markov* on the potential gives some good level inside the window; the *kinetic* term lets us transport that good level to any chosen target inside the window without spoiling the bound.

We need both bounds. Without the kinetic bound (only the potential bound), Markov alone gives "most levels are near B_1" but does not let us point to a specific $\widehat{\rho}$ close to a chosen ρ. Without the potential bound (only the kinetic bound), we know the path is slow but have no anchor saying it ever comes close to B_1.

Choosing the window. We pick $L \asymp \sqrt{\delta_T}$ around the target value $\rho_\delta := 1 - \kappa\sqrt{\delta_T}$. This window is short enough that $\widehat{\rho}$ lands within $O(\sqrt{\delta_T})$ of 1 — needed for the transfer step — and long enough that the abstract lemma still delivers $d^2 \le C \cdot \delta_T$:

$$\boxed{\exists\, \widehat{\rho} \in [\rho_\delta - c_0\sqrt{\delta_T},\, \rho_\delta] \quad \text{with} \quad d_{\mathcal{X}}([F_{\widehat{\rho}}],[B_1])^2 \;\le\; C\,\delta_T(\Omega).}$$

Where does the radial trace lemma fit? The FJ inequality directly delivers symmetric-difference control, but the kinetic estimate in Section 3 is written in terms of a radial quantity on $\partial^* E_\rho$ (how far each boundary point is from the sphere of radius ρ around z_ρ). The radial trace lemma is the geometric identity that converts the two FJ-controlled quantities (symmetric difference + normal oscillation, both D_I-controlled) into the radial form that the kinetic estimate consumes. It's a bridge, not a standalone idea: it makes Sections 2 and 3 compatible, so they can be added.

5. Transfer to Ω, square at the end

We now have $d_{\mathcal{X}}([F_{\widehat{\rho}}],[B_1])^2 \le C\delta_T$ and $\widehat{\rho} \in [1 - O(\sqrt{\delta_T}), 1]$. Two consequences:

(a) Unrescale: $\mathcal{A}(E_{\widehat{\rho}}) \le d_{\mathcal{X}}([F_{\widehat{\rho}}],[B_1]) \le \sqrt{C\delta_T}$.

(b) Boundary layer is thin: $|\Omega \setminus E_{\widehat{\rho}}| = \omega_n(1 - \widehat{\rho}^n) \le C\sqrt{\delta_T}$.

Triangle inequality on Fraenkel asymmetry:

$$\mathcal{A}(\Omega) \;\le\; \mathcal{A}(E_{\widehat{\rho}}) + \frac{2|\Omega \setminus E_{\widehat{\rho}}|}{|\Omega|} \;\le\; C\sqrt{\delta_T(\Omega)}.$$

Squaring at the very end preserves the sharp exponent:

$$\boxed{\mathcal{A}(\Omega)^2 \;\le\; C(n,R)\,\delta_T(\Omega).}$$

Closing. Bounded reduction (BDV Lemma 5.3) removes the R dependence, giving $\mathcal{A}(\Omega)^2 \le C(n)\,\delta_T(\Omega)$ for all Ω. Kohler–Jobin's inequality converts this to

$$\boxed{\delta_{FK}(\Omega) \;\ge\; c_{FK}(n)\,\mathcal{A}(\Omega)^2 \qquad \text{for every open } \Omega \subset \mathbb{R}^n \text{ of finite measure.}}$$

6. One-page recap

Identity	:	$\delta_T(\Omega) = \int (D_H + D_I)\, w\, d\rho$.
What is D_I?	:	non-ballness of E_ρ.
What is D_H?	:	non-uniformity of $\lvert\nabla u\rvert$ on $\partial^* E_\rho$.
D_I **controls**	:	potential $d_{\mathcal{X}}([F_\rho],[B_1])^2$ (via Fusco–Julin).
D_H **controls**	:	kinetic $\lvert\dot F_\rho\rvert^2_{\mathcal{X}}$ (via metric first variation).
Bridge	:	radial trace lemma makes potential and kinetic compatible.
Sum	:	$\int (d^2 + \lvert\dot F\rvert^2)\, w\, d\rho \le C\,\delta_T$.
Metric trace	:	$\exists\, \widehat{\rho} \in [\rho_\delta - c_0\sqrt{\delta_T},\, \rho_\delta]$ with $d^2 \le C\,\delta_T$.
Transfer	:	$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\widehat{\rho}}) + \lvert\Omega \setminus E_{\widehat{\rho}}\rvert/\lvert\Omega\rvert \le C\sqrt{\delta_T}$.
Square	:	$\mathcal{A}(\Omega)^2 \le C\,\delta_T(\Omega)$.
KJ	:	$\delta_{FK}(\Omega) \ge c_{FK}(n)\,\mathcal{A}(\Omega)^2$.

The proof in one sentence. The torsion deficit δ_T decomposes exactly into a kinetic+potential integral along the level-set foliation, both terms small force the foliation, viewed as a path in the shape space $\mathcal{X}$, to pass close to the unit ball at a specific level $\widehat{\rho} \approx 1 - O(\sqrt{\delta_T})$; transferring back to Ω and squaring gives the sharp $\mathcal{A}(\Omega)^2 \lesssim \delta_T$.